

Part 1: "Pre-Startup Information and Checks"

Note: Most measurements can be compared to the Operations & Maintenance Manual or the factory assembly record form number QAF-262. All readings are assumed to be in English units. If metric is used, please label accordingly.

Reseller: _____ Date: _____

Compressor Model: _____ Compressor S/N: _____

Reseller's Unit #: _____ Customers Unit #: _____

Driver Make: _____ Driver Model: _____ Driver S/N: _____

Driver Rated HP: _____ Driver Rated RPM: _____ Variable Frequency Drive (VFD)? Yes _____ No _____

Customer: _____ Customer Contact: _____ Customer Phone#: _____

Reseller Contact: _____ Reseller Phone #: _____

Siemens Energy Rep. Contact (if applicable): _____ Siemens Energy Rep. Phone #: _____

Unit Location(Address): _____ State: _____ Zip: _____

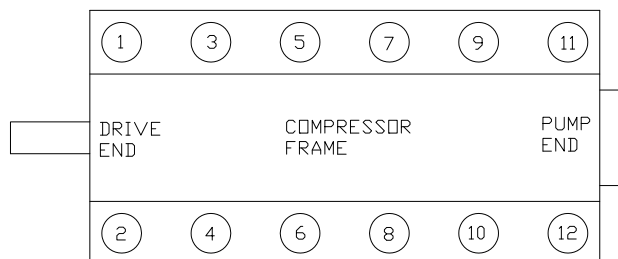
Frame Oil Make: _____ Frame Oil Grade: _____

Cylinder Oil Make: _____ Cylinder Oil Grade: _____

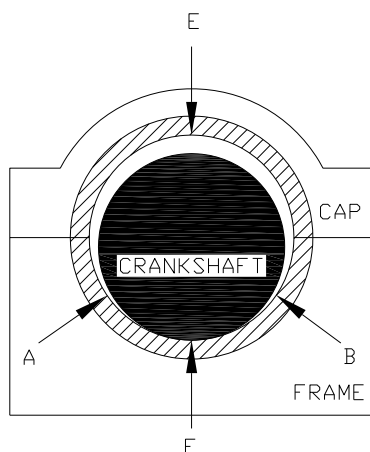
Compressor Application (booster, storage, transmission, EOR, gathering, lift, etc.) _____

1) Record the pull-down readings on the frame hold down bolts.

Frame Foot Bolt #	1	2	3	4	5	6	7	8	9	10	11	12
Pull- down Reading												



2) Check the main bearing clearances. Use a .0015" feeler to check at positions A, B, and F. For A and B positions you should start at the bearing split line and move downward approximately 1 to 1-1/2. You are only looking for the presence of clearance at these locations and can record the results as pass or fail in the table below. At point F you should not be able to insert a .0015" feeler and can record as pass or fail. Select a feeler that fits best at position E and record that number in the table below. **Note: There may be some bearings that cannot be accessed practically- leave those spaces blank.**



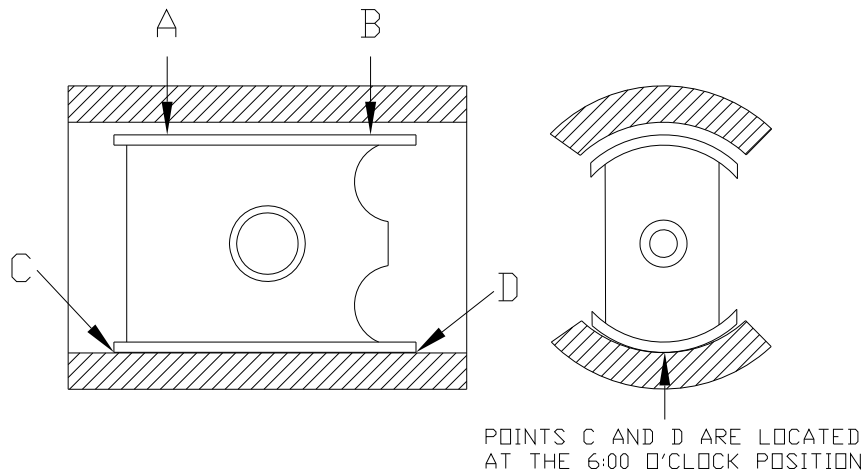
Note: Main Bearing #1 starts from the oil pump end

Location	A	B	E	F
Bearing # 1				
Bearing # 2				
Bearing # 3				
Bearing # 4				
Bearing # 5				
Bearing # 6				

3) Piping and bottle supports have been checked to ensure they are not inducing strain on the compressor cylinders or frame. Yes_____

If strain has been found, where was it and what was done to fix it?

4) Check the crosshead clearances. **Note: Use a .0015" feeler to probe at the bottom locations C and D and record Pass or Fail. The .0015" feeler can go in as much as 1/2" and still pass. Select a feeler that fits best at locations A and B and record those readings in the table below.**



Throw #	A	B	C	D
1				
2				
3				
4				
5				
6				

5) Check piston rod run out. **Note: Sweep the piston from outer end to the frame end when taking these readings. The dial indicator should be as close as possible to the packing flange during these checks.**

Throw #	Vertical Runout	Horizontal Runout
1		
2		
3		
4		
5		
6		

6) Check piston end clearances. **Note: Do not use lead wire- feelers only.**

7) Are there cylinder inboard or outboard supports installed? Check each that apply.

8) What are the “cold” coupling alignment readings?

0.000

0. _____

18) Is the low oil pressure switch tubed to the main bearing header (at the flywheel end)? Yes

- 19) Crankcase breather(s) are installed and open? Yes _____ Note: If the unit is inside of a building the frame vents need to be piped to the outside.
- 20) All vent and drain lines (Frame, VVCP, Packing, Distance Piece, Unloader actuators & stem seals) are installed, properly located, and unrestricted. Yes _____
- 21) The cylinder and packing lubrication has been primed for running Yes _____
- 22) Cylinder lubrication no flow shuts the unit down? Yes _____ No _____
- 23) Lubricator pumps are set to maximum flow? Yes _____
- 24) Is there a vibration switch located on the frame? Yes _____ No _____ If not a switch needs to be installed.
- 25) The vibration switch shuts the unit down? Yes _____ No _____
- 26) All safety shutdowns on the package have been tested? Yes _____
- 27) Pre-Lube pump has been installed and tested. Yes _____ Test pressure? _____ (psig)
- 28) All gas line connections have been blown out prior to hook up to the compressor to ensure they are clean? Yes _____
- 29) Are there a startup screen(s) installed? Yes _____ No _____ If so is there a way to measure pressure drop across the screen in case it becomes plugged?
Yes _____ No _____

Where are the screen(s) located? _____

- 30) All fasteners with gasketed joints have been rechecked for proper torque? Yes _____ These can include, but are not limited to outer head VVCP nuts, all frame and distance piece cover bolting, valve cover bolting, front cover bolts, any oil piping fasteners, piston rod packing nuts, etc.. List any that were not properly torqued.

- 31) Have all clearance pockets been positioned according to the original performance? (if equipped) Yes _____ N/A _____

- 32) The compressor has been barred over at least one full revolution to ensure there are no obstructions? Yes _____

- 33) List any issues found during the Pre-Startup Checks and what was done to resolve those issues:

Part 2: "Startup Information and Checks"

Note: Most measurements can be compared to the Operations & Maintenance Manual or the factory assembly record form number QAF-262. All readings are assumed to be in English units. If metric is used, please label accordingly.

Startup Date: _____

Customer Contact: _____ Customer Phone #: _____

Compressor S/N: _____

Reseller Contact: _____ Reseller Phone #: _____

Siemens Energy Rep. Contact (if applicable): _____ Siemens Energy Rep. Phone #: _____

1) Is the frame pre-lube pump running prior to starting the compressor? Yes _____ No _____ What is the pre-lube pressure? _____ (psig)

2) Main compressor oil pump pressure came up immediately after starting the compressor? Yes _____

What is the main oil pump pressure? _____ (psig)

3) What is the ambient temperature? _____ (°F)

4) What was the oil temperature just prior to start up? _____ (°F)

5) What was the oil temperature when the unit was loaded? _____ (°F)

6) What is the oil filter differential pressure? _____ (psid)

7) Were there any gas, oil, or coolant leaks? Yes _____ No _____ If so explain where and what was done to repair the affected area.

Note: All values are assumed to be taken basis full load conditions for steps 8-13.

8) Does the original performance fall in line with the actual running conditions? Yes _____ No _____ List the running conditions below.

Cylinder #	Cylinder Serial Number	Suction Pressure (psig)	Suction Temperature °F	Discharge Pressure (psig)	Discharge Temperature °F
1					
2					
3					
4					
5					
6					
RPM		Flow Rate MMSCFD			

9) If the compressor is equipped with main bearing RTD's, record the temperatures below. **Note: Main Bearing #1 starts from the oil pump end of the compressor.**

Main Bearing #	Temperature °F
1	
2	
3	
4	
5	
6	

10) Cylinder jacket water cooling temperature (if equipped)

Cylinder #	Temperature In °F	Temperature Out °F
1		
2		
3		
4		
5		
6		

9) If the compressor is equipped with packing RTD's, record the temperatures below. **Note: Packing #1 starts from the oil pump end of the compressor.**

Packing #	Temperature °F
1	
2	
3	
4	
5	
6	

11) What is the differential pressure across the suction screen(s) _____ psig Location _____
 _____ psig Location _____
 _____ psig Location _____

12) Heat exchanger oil temperature IN _____ (°F) Heat exchanger oil temperature OUT _____ (°F)

13) Heat exchanger coolant temperature IN _____ (°F) Heat exchanger coolant temperature OUT _____ (°F)

14) The cylinder lubrication no flow shuts the unit down? Yes _____

15) The vibration switch(s) shut the unit down? Yes _____

16) Is the low frame oil pressure switch is working? Yes _____

What is the shut down pressure? _____ psig

17) Cycle pin indicators in the cylinder lubrication system cycle (if equipped) Yes _____ N/A _____

18) All coolant flow indicators are working? (if equipped) Yes _____ N/A _____

19) What is the supply pressure to the pneumatic valve unloaders or pockets? (if equipped) _____ (psig)

Which is being used to operate the devices? Air _____ Process Gas _____

20) Has a "hot" coupling alignment been done? Yes _____ No _____ (should not exceed .005" TIR)

0.000

0.000

0. _____ Face 0. _____ 0. _____ Rim 0. _____

0. _____

0. _____

21) Discharge bottle chocks have been adjusted to support the weight of the bottle after unit has been heat soaked? Yes _____

22) Lube pump(s) are set to maximum flow? Yes _____

23) What is the master divider block cycle pin indicator cycle time? _____ (seconds)